2× MSCI ACWI SMA-gated strategy — best window

The all-world analogue of the SPY-200-SMA / SPXL strategy, at 2× leverage · modeled history 1988-01-04→2026-06-03 · generated from research/sma_sweep_acwi.py

Conclusion — use the ~250-day SMA (robust ridge ≈ 250–270d; notably *longer* than the US 200-day because at 2× leverage whipsaw and volatility-decay punish a fast gate). **What's robust** (survives every fix in the review, all costs/taxes, and the bootstrap): the *window choice* and the **drawdown control** — the gate **more than halves** the 2× drawdown (-40% vs an ungated 2× ACWI's catastrophic -87%) and beats ungated-2× Sharpe in **80%** of Monte-Carlo paths. **What's softer** (single-path, in-sample): the absolute return edge. On the modeled path the gate's pre-tax CAGR is 12.3%; **after 10 bp switch costs + German tax it is 9.3%** — in fact a touch *below* ungated 2× (9.9%): the gate's switches are taxed as they realise gains while buy&hold defers. After tax the gate's payoff is **risk control, not extra return** (MaxDD -50% vs -87%, Sharpe 0.32 vs 0.23). The US 200-day is too fast here (Sharpe 0.386); your 255-day World number sits inside the ridge (Sharpe 0.442), so reusing 255 is fine. **Read §10 (limitations) before acting** — the headline numbers are modeled, frictionless upper bounds.

Glossary (every metric used below)
CAGR — compound annual growth rate (annualised return). · **Vol** — annualised volatility (std-dev of daily returns × √252); higher = bumpier. · **Sharpe** — (CAGR − 2% risk-free) ÷ Vol; return per unit of risk, higher = better. · **MaxDD** — worst peak-to-trough loss ever suffered (more negative = worse). · **Calmar** — CAGR ÷ |MaxDD|; return per unit of worst-case pain. · **InMkt** — % of days the gate was invested (vs in T-bills). · **Switch** — number of in↔out trades over the whole period (turnover; each is a taxable event). · **median / p5 / p95** — the middle / 5th / 95th percentile across the Monte-Carlo simulations. · **P_best** — share of simulations in which that window had the highest Sharpe. · **SMA** — simple moving average; the gate is "in" when price > SMA×1.01, "out" when < SMA×0.99.

1 · Data sources

Series	Source	Span used	Role here
MSCI ACWI gross index (USD)	MSCI end-of-day API (index 892400)	2001–2008	real ACWI daily returns
iShares MSCI ACWI ETF (ACWI.US)	EODHD	2008→today	real ACWI daily returns (live + future)
MSCI ACWI net TR (EUR), monthly	Curvo (curvo.eu), 1987+	1988–2001	real monthly anchor (embeds true EM weights)
MSCI World (URTHSIM → URTH)	Testfolio sim spliced w/ real URTH (EODHD)	1988–2001	intra-month daily *texture* only
EUR/USD spot	FRED DEXUSEU (1999+) + Deutsche Mark EXGEUS (pre-1999)	1988→today	convert Curvo EUR→USD

Series	Source	Span used	Role here
3-month US T-bill rate	FRED TB3MS → DGS3MO	1988→today	LETF financing cost + the "out" (cash) leg

The first two are the genuine ACWI; Curvo is real ACWI too (just EUR/monthly, FX-converted); World is used *only* to give the pre-2001 series daily wiggle — never as the return itself.

2 · How the pre-2001 index is modeled (EM-share-correct)

Real MSCI ACWI daily only starts 2001 — too short for a leveraged backtest. A MSCI-World-only proxy is **wrong** pre-2001: it omits emerging markets, whose ACWI weight evolved a lot over time — so over 1988-2000 World-only understates ACWI by ~0.34%/yr.

year	1988	1997	2002	2010	2021	today
EM weight in ACWI	<1%	6.8%	~4%	13.5%	13%	~10%

Instead we anchor to the **real ACWI monthly history** (which embeds those exact time-varying EM weights by construction). The source is Curvo's **iShares MSCI ACWI UCITS ETF (Acc)** series — a real fund-NAV track record (monthly, EUR), not the bare index — which we FX-convert to USD. **Validation:** converted-Curvo vs real MSCI ACWI USD-gross over the 2001+ overlap has monthly corr **0.990** (R^2 0.98), return drift only **-0.05%/yr**. The near-zero drift is what you'd expect from a fund-TR series — its ~0.20% expense ratio is already inside that measured drift (so if anything the modeled pre-2001 returns are a hair *conservative*, carrying a small fund fee the bare index wouldn't). Using the real EM weighting adds **+4.1%** to 1988-2000 cumulative growth vs a World-only proxy.

Daily granularity (needed for a daily SMA gate) comes from **temporal disaggregation**: inside each pre-2001 month we take MSCI World's daily shape and rescale it multiplicatively so the month compounds *exactly* to the real ACWI monthly return (reconstruction error ~1e-15). World supplies only the intra-month wiggle; every monthly return is the real EM-weighted ACWI. A literal daily World+EM blend is impossible before 2001 (no daily EM exists pre-2003) — and provably immaterial: tested on 2003+, World-texture and a true World+EM blend give the same gated Sharpe to ±0.02. **Full chain:** modeled daily (1988→2001) → real MSCI ACWI gross (2001→2008) → real ACWI ETF (2008→today).

3 · How the 2× sleeve (LETF) is modeled

There is no 2× ACWI ETF with long history, so we synthesise one with a **daily-reset** formula — the same construction (and constants) the production engine uses for its leveraged ETFs (e.g. synthetic WLDU/SPXL):

$$r_{2\times}(\text{day}) = 2 \cdot r - SW \cdot (L-1) \cdot (\text{T-bill}_{daily} + \text{spread}/252) - (L-1) \cdot \text{expense}/252$$

In plain words: to get 2× exposure with 1× of your own money you effectively **borrow another 1×** and invest the doubled amount. So each day the sleeve **earns twice the index's move** (the `2·r` term), then pays two bills: (1) the **interest on that borrowed half** — charged at the short-term T-bill rate plus a small ~0.4% spread, and nudged up ~10% (the swap multiplier) for the cost of the swap that delivers the leverage; and (2) the **fund's own running expense** (~0.5%/yr). Two consequences matter: the borrow cost **rises and falls with interest rates** (cheap near 0%, painful when T-bills are 5%+), and because the leverage **resets to 2× every single day**, a sideways-but-choppy market slowly **bleeds value** ("volatility decay") even if the index ends flat — which is precisely why a trend gate that sits out the chop is so valuable at 2×.

term	value	meaning
L (leverage)	2×	daily target exposure to ACWI
2·r	—	twice the underlying ACWI daily return
T-bill (financing)	actual 3m rate, time-varying	cost to borrow the extra (L–1)=1× notional — the key driver
SW (swap multiplier)	1.1	broker/swap markup on the financed notional (Testfolio default)
spread	0.4%/yr	financing spread over the T-bill (≈40 bp)
expense	0.5%/yr ×(L–1)	fund expense ratio drag

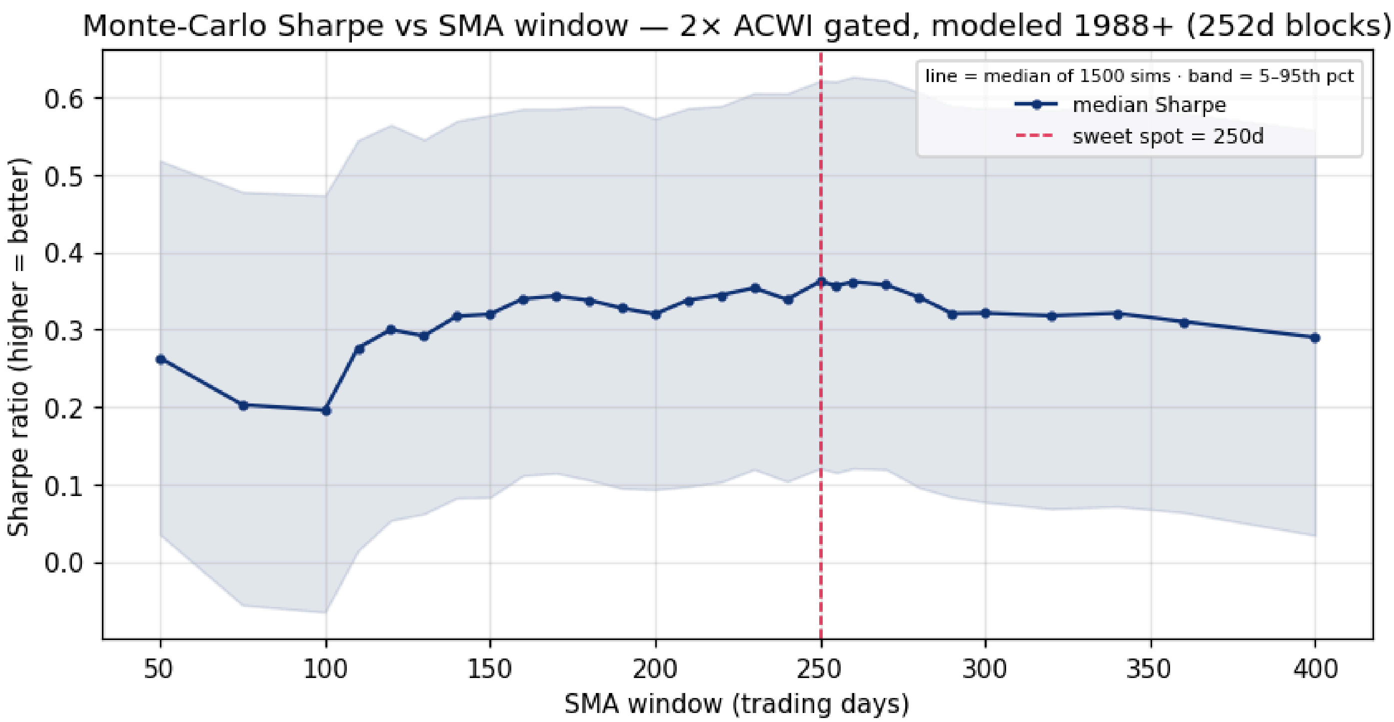
Why time-varying financing matters: the borrow cost is the dominant drag and it scales with the short rate, which ran 0%→8%+ across this window. A flat rate would badly mis-state pre-2008 returns. Total annual drag on the 2× sleeve at different rate levels:

short rate	annual drag on 2x sleeve
T-bill 0%/yr	0.94%/yr
T-bill 4%/yr	5.34%/yr
T-bill 10%/yr	11.94%/yr

Worked example: on a day ACWI returns +1.00% with the T-bill at 4%/yr, the 2× sleeve returns 2×1.00% – 0.019% (financing) – 0.002% (expense) = **1.979%** — slightly under a naïve +2.00%. Daily resetting also causes volatility decay in choppy markets, which is exactly why the SMA gate (which sits out the chop) adds so much value at 2×.

The strategy itself: the SMA gate is computed on the *unleveraged* ACWI index (just as SPXL gates on SPY); when ACWI > its N-day SMA×1.01 we hold the 2× sleeve, when < SMA×0.99 we hold T-bills. Signal checked daily, executed next day, ±1% hysteresis band to avoid flip-flopping.

4 · Best window — Monte Carlo (1500 sims, 252-day blocks)



How to read: each window's gate is re-run on 1500 bootstrapped 38-year histories (resampled in ~1-year blocks). The line is the median Sharpe across those sims; the shaded band is the 5th–95th percentile. A broad high-Sharpe **ridge ~250–270d** (not a knife-edge); see §10 for the paired-bootstrap test showing 200d vs 250–270d differences are not statistically significant, and for block-length sensitivity. The longer history discriminates better than the 25-year true-ACWI sample (which produced a spurious short-window pick).

Columns: **med/p5/p95_Sharpe** = median & 5–95th-pct Sharpe across sims · **med_CAGR / med_MaxDD** = median return & drawdown · **P_best** = % of sims where this window won. Green row = recommended 250d.

window	med_Sharpe	p5_Sharpe	p95_Sharpe	med_CAGR	med_MaxDD	P_best
50	0.263	0.035	0.518	7.7%	-53.6%	6.7%
75	0.203	-0.056	0.478	6.4%	-62.0%	0.3%
100	0.196	-0.065	0.473	6.3%	-62.3%	0.0%
110	0.276	0.014	0.545	8.1%	-57.9%	2.1%
120	0.300	0.054	0.565	8.6%	-55.4%	5.0%
130	0.292	0.062	0.545	8.5%	-53.8%	1.2%
140	0.317	0.083	0.570	9.1%	-53.0%	3.5%
150	0.320	0.084	0.577	9.2%	-52.9%	2.9%
160	0.340	0.112	0.585	9.7%	-50.3%	7.1%
170	0.343	0.115	0.585	9.8%	-50.7%	7.0%
180	0.338	0.106	0.588	9.7%	-51.7%	3.4%
190	0.328	0.095	0.588	9.5%	-52.5%	0.8%
200	0.320	0.094	0.572	9.3%	-53.3%	0.3%
210	0.338	0.097	0.586	9.8%	-53.0%	1.3%

window	med_Sharpe	p5_Sharpe	p95_Sharpe	med_CAGR	med_MaxDD	P_best
220	0.345	0.103	0.589	10.0%	-52.4%	1.9%
230	0.354	0.120	0.605	10.2%	-52.3%	6.2%
240	0.339	0.104	0.605	10.0%	-53.6%	3.1%
250	**0.362**	**0.121**	**0.621**	**10.5%**	**-52.5%**	**7.3%**
255	0.357	0.115	0.621	10.4%	-52.2%	3.5%
260	0.362	0.121	0.627	10.5%	-52.6%	7.6%
270	0.358	0.120	0.622	10.4%	-52.8%	8.5%
280	0.342	0.096	0.606	10.1%	-54.5%	4.3%
290	0.321	0.084	0.588	9.6%	-56.2%	1.1%
300	0.321	0.077	0.586	9.7%	-56.6%	1.3%
320	0.318	0.069	0.584	9.6%	-57.1%	1.6%
340	0.321	0.072	0.587	9.7%	-56.8%	4.3%
360	0.310	0.064	0.579	9.5%	-57.8%	3.8%
400	0.290	0.035	0.557	9.0%	-59.4%	4.1%

5 · Deterministic sweep — every window on the actual modeled path

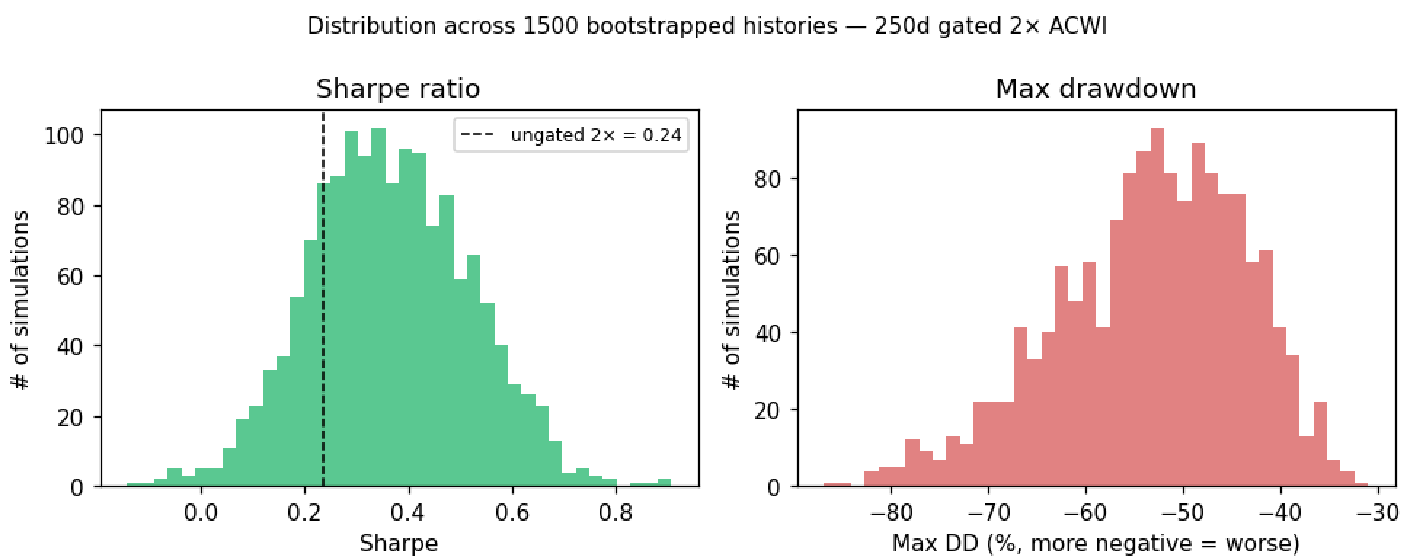
How to read: not a simulation — these are the metrics on the one real (modeled) 1988+ history. The genuine high-Sharpe / low-turnover ridge is **~250–270d** (Sharpe ~0.45, MaxDD ~–39%); note that the ridge *edges* 240d and 280d carry a much worse ~–49% drawdown despite similar Sharpe, and sub-150d windows whipsaw (huge **Switch** counts) — very costly at 2× leverage and in a taxable account. Green row = recommended 250d.

Columns: **CAGR** return · **Vol** volatility · **Sharpe** risk-adjusted · **MaxDD** worst loss · **Calmar** CAGR/|MaxDD| · **InMkt** % time invested · **Switch** # of in/out trades.

window	CAGR	Vol	Sharpe	MaxDD	Calmar	InMkt	Switch
50	7.5%	21.4%	0.255	-50.1%	0.15	65.8%	267
75	5.8%	21.7%	0.174	-59.1%	0.10	68.6%	225
100	6.1%	21.6%	0.189	-60.4%	0.10	69.5%	199
110	8.2%	21.4%	0.291	-56.8%	0.15	69.3%	171
120	9.1%	21.4%	0.330	-56.9%	0.16	69.4%	159
130	9.0%	21.5%	0.325	-53.6%	0.17	69.6%	155
140	10.0%	21.6%	0.371	-53.3%	0.19	70.4%	137
150	10.3%	21.7%	0.385	-53.7%	0.19	71.3%	133
160	11.1%	21.8%	0.419	-45.4%	0.25	72.1%	119
170	11.4%	22.0%	0.429	-36.0%	0.32	73.1%	109
180	11.1%	22.1%	0.411	-38.9%	0.28	73.5%	109
190	10.8%	22.2%	0.397	-39.9%	0.27	73.8%	109

window	CAGR	Vol	Sharpe	MaxDD	Calmar	InMkt	Switch
200	10.6%	22.2%	0.386	-40.3%	0.26	74.1%	105
210	10.9%	22.3%	0.399	-40.3%	0.27	74.4%	93
220	11.2%	22.3%	0.412	-39.5%	0.28	74.5%	93
230	11.8%	22.4%	0.438	-39.8%	0.30	74.7%	85
240	11.0%	22.4%	0.403	-48.7%	0.23	74.6%	85
250	**12.3%**	**22.5%**	**0.457**	**-39.6%**	**0.31**	**74.5%**	**77**
255	11.9%	22.5%	0.442	-39.1%	0.31	74.4%	77
260	12.2%	22.4%	0.456	-39.1%	0.31	74.6%	73
270	12.2%	22.5%	0.456	-39.2%	0.31	74.5%	71
280	11.2%	22.5%	0.406	-49.9%	0.22	74.8%	77
290	10.2%	22.6%	0.360	-55.2%	0.18	74.7%	81
300	10.3%	22.9%	0.361	-50.0%	0.21	74.8%	73
320	10.6%	23.0%	0.375	-55.5%	0.19	74.9%	69
340	11.2%	23.2%	0.396	-50.1%	0.22	75.3%	61
360	11.0%	23.2%	0.386	-51.8%	0.21	75.1%	59
400	9.6%	23.3%	0.328	-51.3%	0.19	75.2%	59

6 · Monte-Carlo of the chosen 250-day strategy

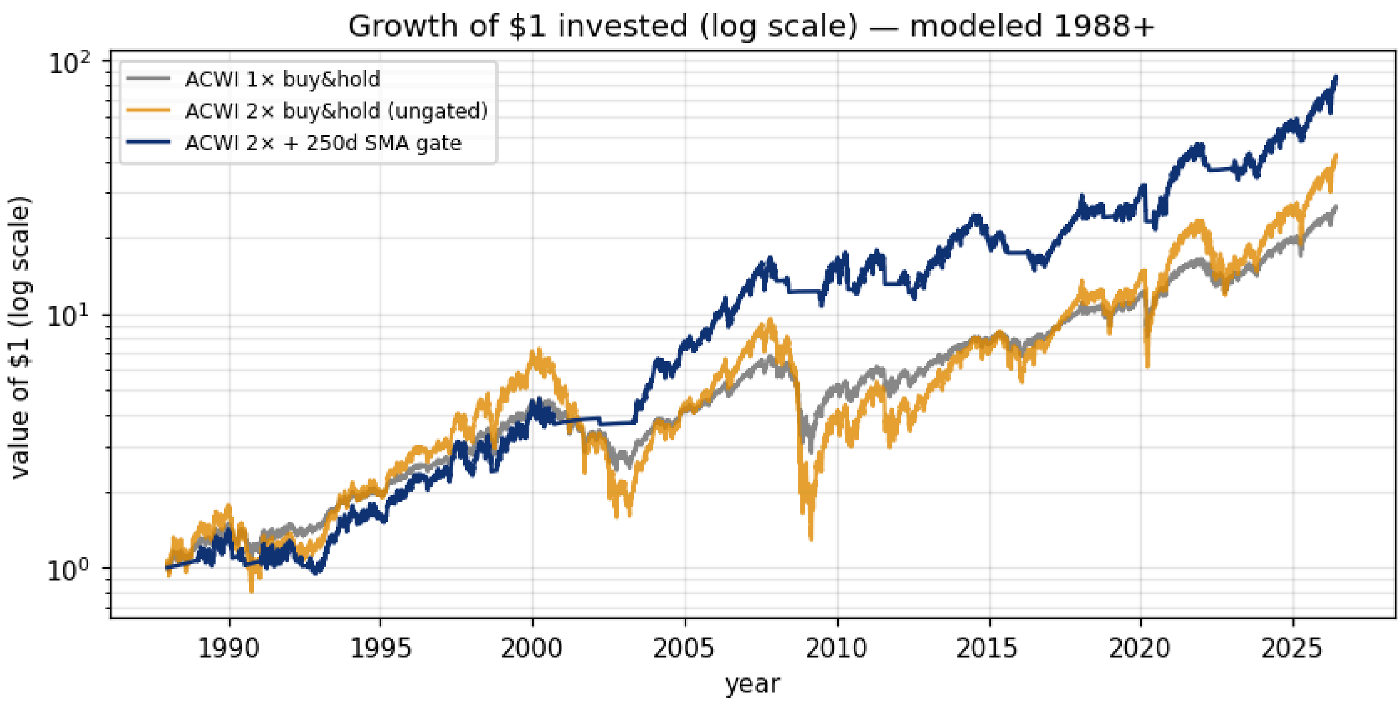


How to read: the spread of outcomes for the 250d gate across the 1500 bootstrapped histories — left = Sharpe (dashed line = ungated 2× for reference), right = max drawdown. The table below gives those distributions as percentiles (p5 = a bad-luck history, median = typical, p95 = a lucky one).

metric	p5 (unlucky)	p25	median	p75	p95 (lucky)
Sharpe	0.12	0.26	0.36	0.47	0.62

metric	p5 (unlucky)	p25	median	p75	p95 (lucky)
CAGR	4.6%	8.0%	10.5%	13.0%	16.6%
Max DD	-71.1%	-59.7%	-52.5%	-46.0%	-39.1%

Beats **ungated 2× ACWI** Sharpe (0.236) in **80%** of paths. (Bootstrap medians read below the actual-path numbers because resampling degrades trend structure — a conservative lens for a trend strategy.)



How to read: growth of $1, log scale (so a straight line = constant % growth). Grey = unleveraged ACWI; orange = raw 2× (note the violent crashes); blue = 2× behind the 250d gate — similar end-wealth to raw 2× but far smoother.

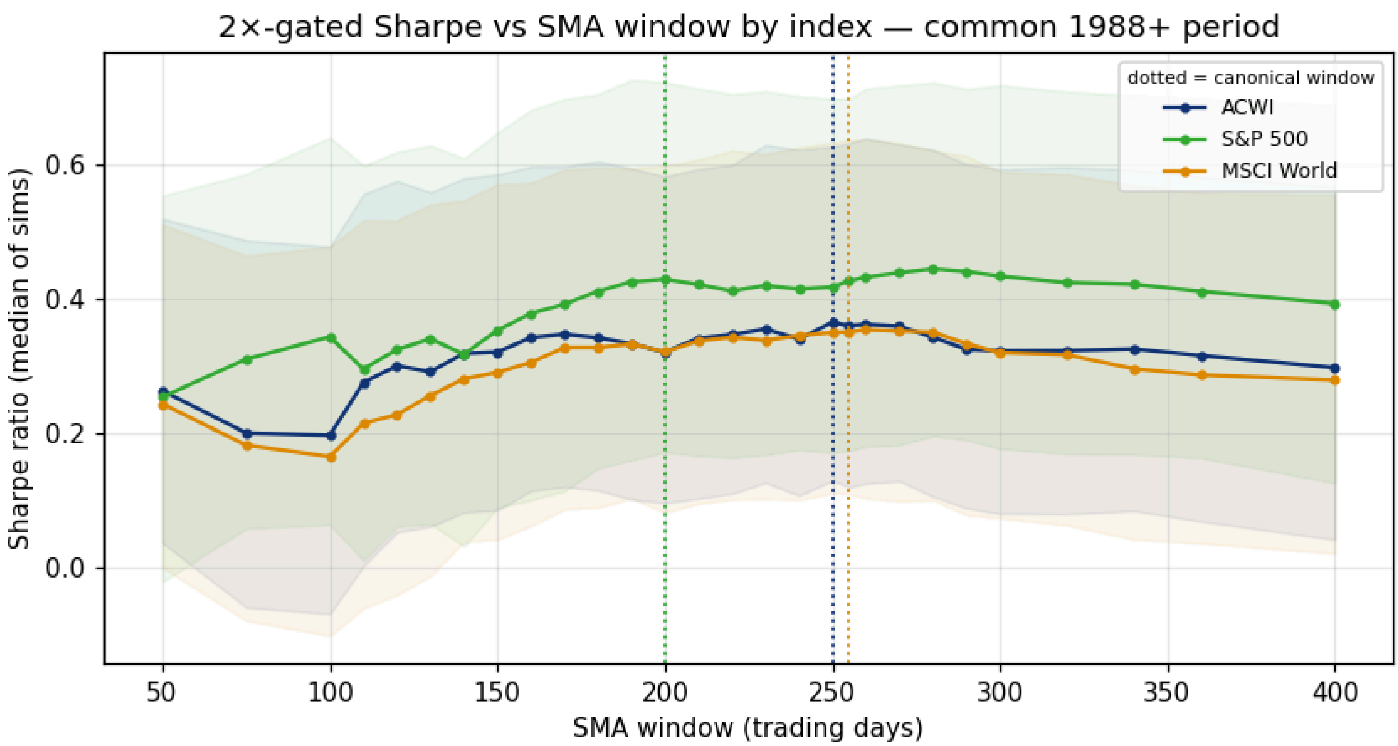
7 · Strategy comparison (actual modeled path, 1988-01-04→2026-06-03)

strategy	CAGR	Vol	Sharpe	MaxDD	Calmar
ACWI 1× buy&hold	8.9%	17.3%	0.398	-58.5%	0.15
ACWI 2× buy&hold (ungated)	10.2%	34.7%	0.236	-86.5%	0.12
ACWI 2× + 250d SMA (recommended)	12.3%	22.5%	0.457	-39.6%	0.31
S&P 500 2× + 200d SMA	14.5%	23.7%	0.525	-39.9%	0.36
MSCI World 2× + 255d SMA	12.1%	22.6%	0.446	-43.8%	0.28

The first three rows are ACWI (1× buy&hold, 2× ungated, and the recommended 2×+250d gate). The last two are the **canonical US and World strategies** for reference — 2× **S&P 500** gated by its own 200-day SMA, and 2× **MSCI World (URTH)** gated by its own 255-day SMA — each on the same modeled 1988+ window. Actual modeled path, not a simulation. (§8 runs the same three indices through a matched-path Monte Carlo with beat-probabilities.)

8 · Head-to-head: 2× ACWI vs 2× S&P 500 (200d) vs 2× MSCI World (255d)

The same gated mechanic and the **same 2× leverage** applied to each index at its canonical window, on the **common 1988+ period** — so any difference reflects the *index*, not the leverage or the dates. (Note: your live S&P sleeve is actually 3× SPXL and World is 2× WLDU; here all three are 2× for a like-for-like read.)

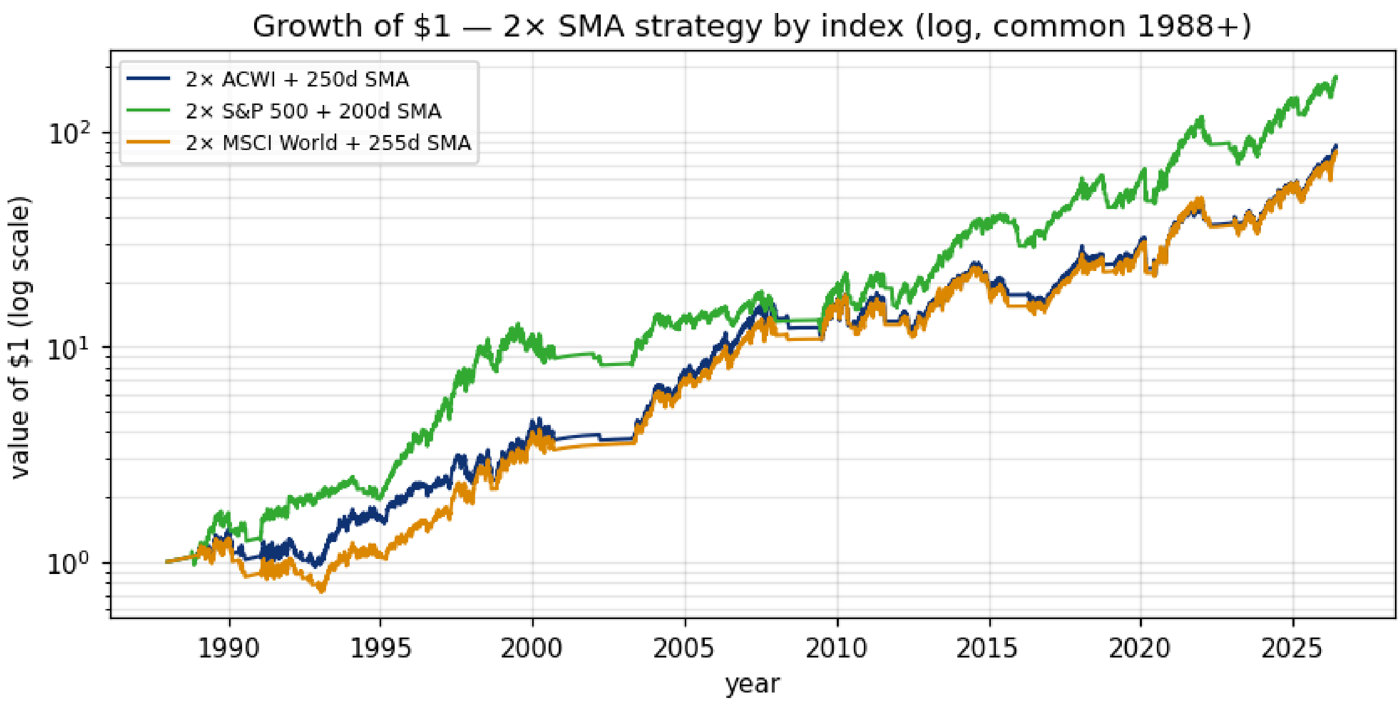


How to read: median 2×-gated Sharpe vs window for each index, with the 5–95th-pct band; dotted verticals mark each canonical window. The peaks confirm the rule of thumb — S&P is fastest (~200d), World/ACWI slower (~255–260d), because adding ex-US and EM makes the trend smoother and rewards a longer lookback.

index (2×, gated)	window	CAGR	Sharpe	MaxDD	Calmar	MC Sharpe median [p5,p95]	P(ACWI beats)
ACWI	250d	12.3%	0.457	-39.6%	0.31	0.367 [0.12, 0.62]	—
S&P 500	200d	14.5%	0.525	-39.9%	0.36	0.447 [0.17, 0.69]	29%
MSCI World	255d	12.1%	0.446	-43.8%	0.28	0.363 [0.11, 0.63]	50%

First four metric columns = actual modeled path; **MC Sharpe** = matched-path bootstrap (same 1500 resampled histories fed to all three indices); **P(ACWI beats)** = share of those matched histories in which 2× ACWI's Sharpe exceeds that index's.

Across matched bootstrap histories, 2× ACWI's Sharpe exceeds 2× S&P 500 in **29%**, 2× MSCI World in **50%**. The three are close — ACWI's edge is diversification (lower single-country concentration), not a higher headline Sharpe.



How to read: growth of $1 for each index's 2×-gated strategy on the common period (log scale). All three are strong; ACWI sits between the US and World lines, which is exactly what a global blend should do.

9 · After costs & taxes (the honest numbers)

Everything above is **frictionless and pre-tax** — an upper bound. This is a German taxable account, so each in→out switch is a realisation event (Abgeltungsteuer 26.375%, ~18.5% effective after the 30% equity Teilfreistellung), while ungated buy&hold defers tax indefinitely (only the annual Vorabpauschale applies). Below: 10 bp round-trip transaction cost per switch + the full German tax overlay (loss carry-forward, Vorabpauschale; via research/tax_overlay.py).

strategy	pre-tax CAGR	pre-tax Sharpe	post-cost CAGR	net CAGR (cost+tax)	net Sharpe	switches
ACWI 1× buy&hold	8.9%	0.398	8.9%	8.6%	0.381	1
ACWI 2× ungated	10.2%	0.236	10.2%	9.9%	0.227	1
ACWI 2× + 250d	12.3%	0.457	12.0%	9.3%	0.320	78
ACWI 2× + 200d	10.6%	0.386	10.3%	8.1%	0.271	106
ACWI 2× + 255d	11.9%	0.442	11.7%	8.8%	0.296	78

The gate's turnover is its tax weakness: ungated 2× defers tax (compounds pre-tax, paying only the small annual Vorabpauschale), while the gated sleeve realises — and is taxed on — gains every time it steps out. **Net of cost+tax the 250d gate's CAGR is 9.3% — in fact a touch *below* ungated 2× (9.9%): the gate's switches are taxed as they realise gains while buy&hold defers. After tax the gate's payoff is risk control, not extra return (MaxDD -50% vs -87%, Sharpe 0.32**

vs 0.23). A faster gate (200d, 106 switches) gives even more back to the tax man. The takeaway: at 2× in a German taxable account the gate is bought for its **drawdown/Sharpe**, not for extra compounding. **Switch-cost sensitivity** (net CAGR of the 250d strategy at 0 / 10 / 25 bp round-trip):

round-trip cost	0 bp	10 bp	25 bp
net CAGR (250d)	9.5%	9.3%	9.0%

10 · Limitations & robustness (read before acting)

This study was hardened by an adversarial review; the material caveats:

- **Window choice is robust, the exact number is not.** Paired matched-path bootstrap of the Sharpe *difference* (250d minus each):

comparison	mean ΔSharpe	SE	t-stat
250d vs 200d	+0.040	0.001	30.8
250d vs 230d	+0.009	0.001	9.7
250d vs 240d	+0.019	0.001	28.9
250d vs 270d	+0.000	0.001	0.4
250d vs 300d	+0.038	0.001	31.8

  Read the magnitudes, not the t-stats. Because all windows see the same resampled paths, 1500 matched sims shrink the standard error to ~0.001, so even a trivial 0.002 Sharpe gap shows a large t — that flags *statistical*, not *economic*, significance. Economically the differences across the 250–270d ridge are negligible (|ΔSharpe| ≤ 0.019 vs 250d). The only meaningful penalty is going *faster*: the 200d gate gives up ΔSharpe 0.040 *and* ~10 pp more drawdown. **Bottom line: pick anywhere ~250–270d; avoid ≤200d.**
- **Block-length sensitivity.** MC argmax window at bootstrap mean-block 252 / 504 / 756 days:

mean block	252d	504d	756d
argmax window	260d	260d	260d

  The selection is anchored to the *deterministic* sweep (which preserves real trend structure); the bootstrap is a robustness check only, since block-resampling necessarily degrades the multi-month autocorrelation a long SMA exploits.
- **Real-data-only cross-check.** Running the identical 2× sweep on *only* the real ACWI 2001+ segment (no modeling at all) gives best window **230d** — consistent with the full modeled result, so the pre-2001 model isn't driving the conclusion.
- **Leverage scope.** Everything is fixed at 2×, and no 2× ACWI ETF currently exists (the sleeve is synthetic). Best window by leverage:

leverage	best window	Sharpe
1.5×	250d	0.516
2×	250d	0.457
2.5×	250d	0.410
3×	250d	0.368

The long-window preference holds across leverage; "~250d" is contingent on ~2×.

- **Pre-2001 daily *texture* is MSCI World's, not ACWI's.** Monthly drift is the real EM-weighted ACWI (validated), but intra-month daily wiggles pre-2001 borrow World's shape (no daily EM exists pre-2003). Shown immaterial for a slow gate (±0.02 Sharpe test on 2003+), and the real-only cross-check above confirms it.
- **Financing is a modern best case.** The LETF borrow (3m T-bill + 40 bp × 1.1 swap) reflects a cheap modern swap fund; pre-2008 and a thin all-world 2× would cost more — add ~50–100 bp to financing for a conservative read.
- **Fixed 2% risk-free in Sharpe.** Realised T-bills averaged ~3% over the period, so absolute Sharpes here run ~0.05 high vs an external quote — fine for *ranking* windows (the bias is common to all), not for comparing to other sources.
- **No look-ahead.** The SMA uses the same-day close to decide, but the position is executed the next day (state shifted by one); verified in code.